Inference-Time Diversity in RL-Trained Lean Theorem Provers

A Diagnostic Study

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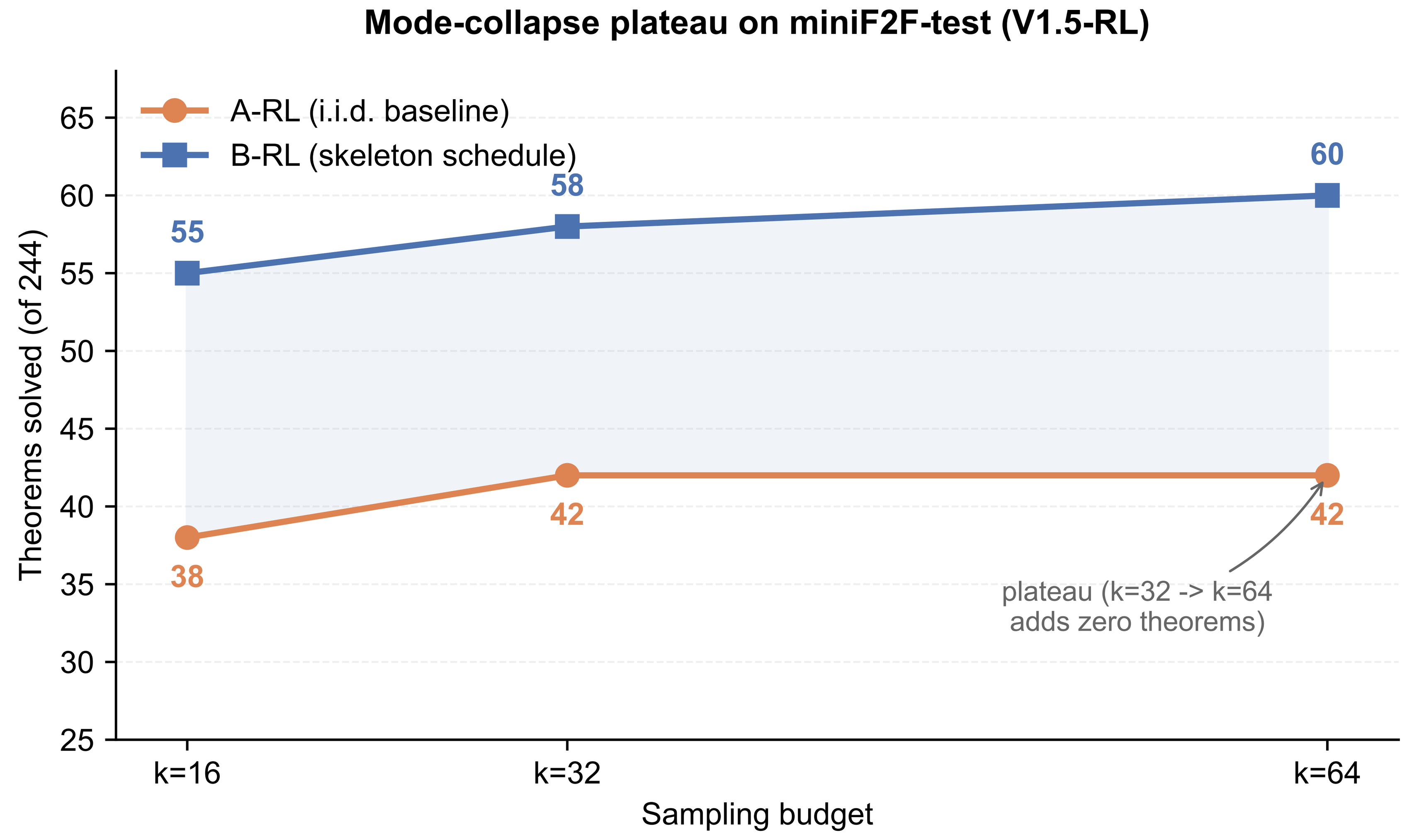
TL;DR *RL-trained Lean theorem provers run out of ideas as you sample more. A fixed schedule of 15 tactic skeletons closes the gap, and the effect is RL-specific.*

Setup

Models (all 7B): primary within-architecture analysis on **DeepSeek-Prover-V1.5-RL** paired with non-RL **V1.5-Base**; cross-model replication on RL-trained **DeepSeek-Prover-V2-7B** (reasoning-mode) and SFT-trained **Goedel-Prover**.
Benchmark: miniF2F-test (244 Lean 4 / Mathlib theorems).
Schedule: 15 hand-picked tactic skeletons (`simp`, `intro`, `constructor`, `aesop`, ...) $\times$ 8 NL goal hints = 120-slot grid. Budgets $k \in \{16, 32, 64\}$ take the first k slots.
Decoding: temperature 0.6, top-p 0.95; max 1024 tokens (V1.5 / Goedel completion-mode) or 8192 tokens (V2 reasoning-mode). **Verifier:** `lake env lean --json`, 120 s timeout, 8-way sharding by $i \bmod 8$.

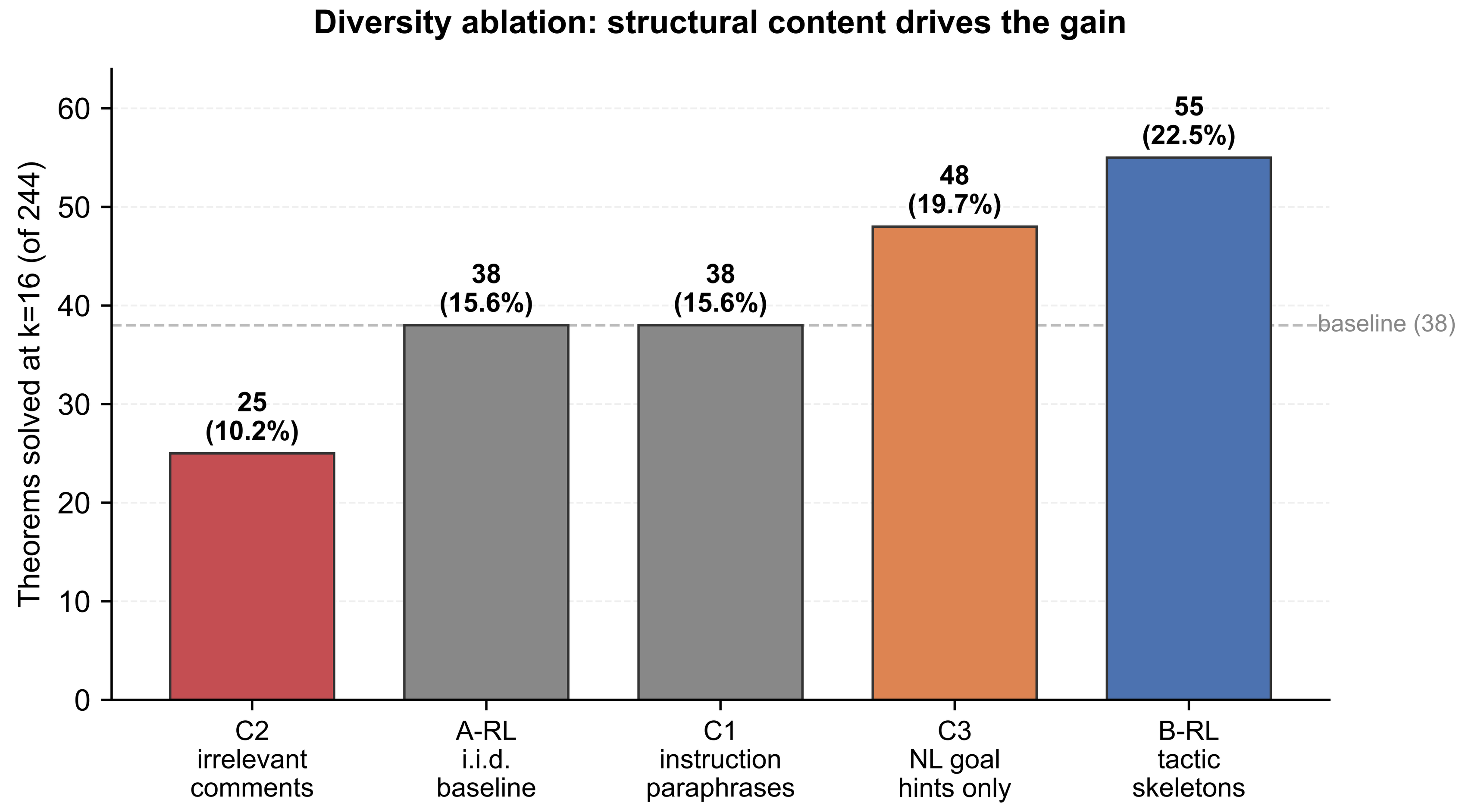
Mode collapse: A-RL plateaus at $k{=}32$



	$k{=}16$	$k{=}32$	$k{=}64$
A-RL (i.i.d.)	38/244 (15.6%)	42/244 (17.2%)	42/244 (17.2%)
B-RL (skeleton)	**55** (22.5%)	**58** (23.8%)	**60** (24.6%)
Δ absolute	$+17$	$+16$	$+18$
Δ relative	$+44.7\%$	$+38.1\%$	$+42.9\%$

Across-seed: mean $\Delta = +12.3 \pm 4.2$ theorems at $k{=}16$ ($n{=}3$ seeds, sign positive every seed).

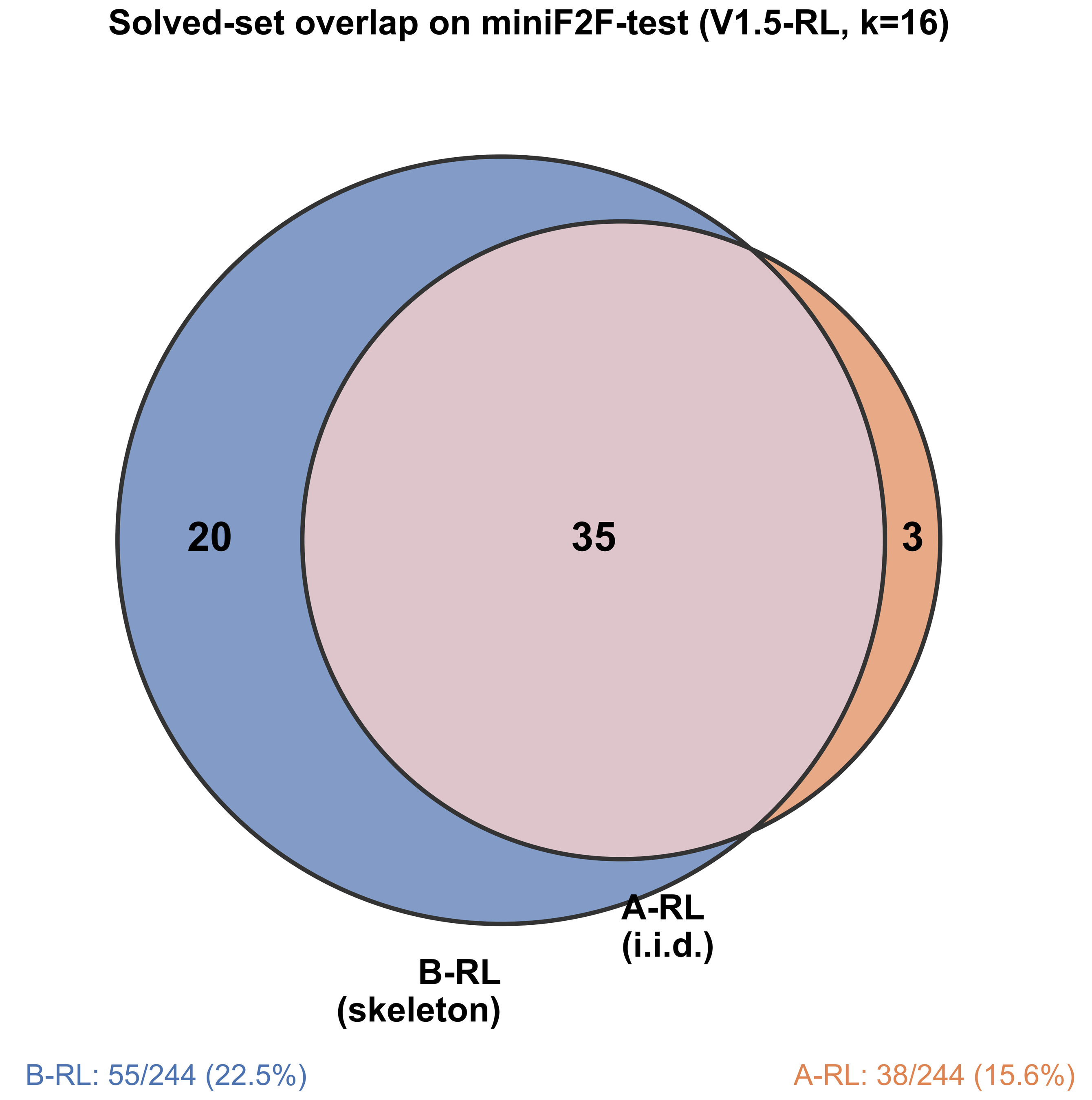
Diversity ablation: structural content drives the gain



C2 < A-RL = C1 < C3 < B-RL. Irrelevant tokens hurt; paraphrase ties the baseline; **C3 (NL hint only, no tactic prefix) alone closes 58% of the A→B gap**; tactic skeletons go furthest.

The RL-collapsed policy doesn't need to be *over-ridden* by a forced first tactic — just *re-conditioned* on a different prompt prefix, and a different head of its narrow distribution becomes accessible.

Solved-set overlap (V1.5-RL, $k{=}16$)



B-RL nearly Pareto-dominates A-RL: **20** theorems are solved only by B-RL, **3** only by A-RL, **35** by both.

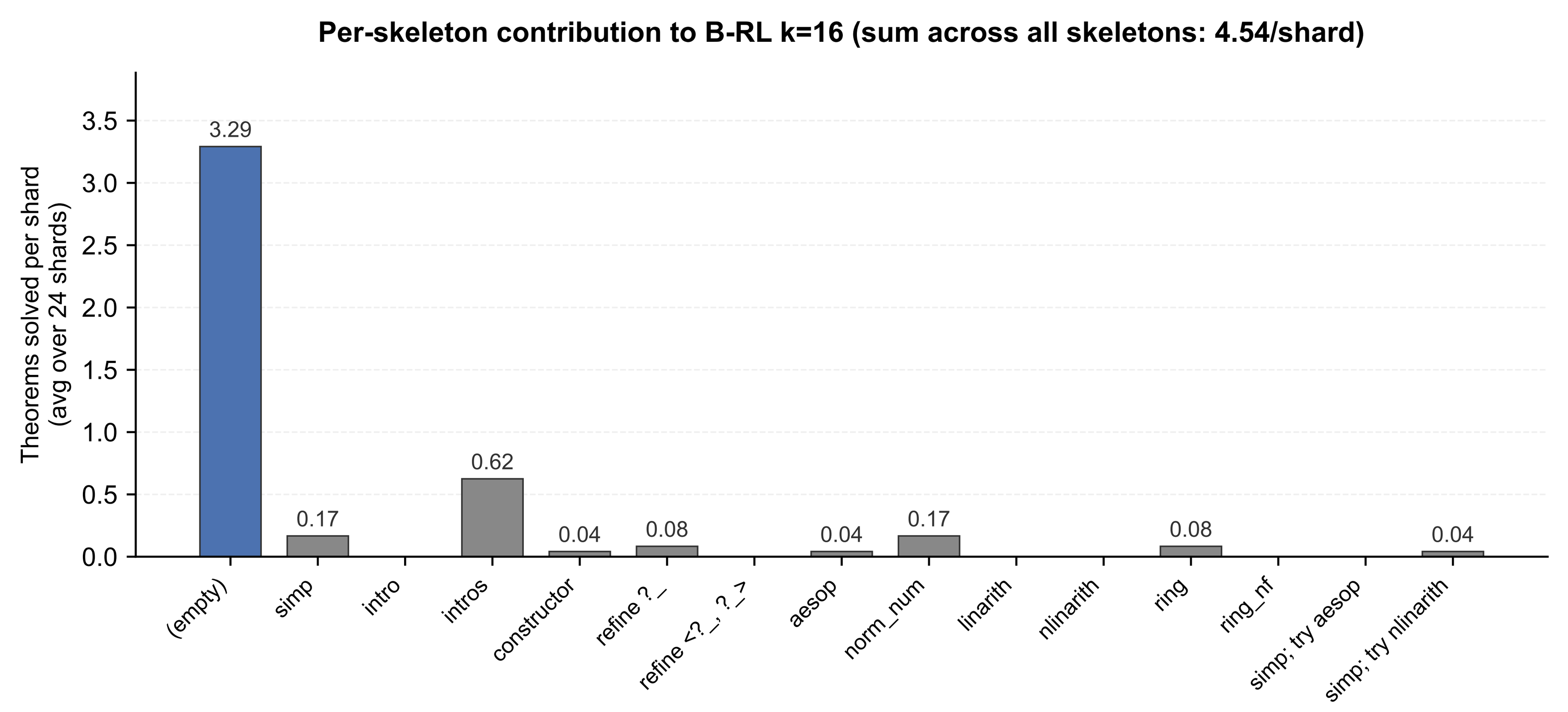
Phenomenon is RL-specific

DeepSeek-Prover-V1.5-Base (same model, no RL fine-tune) proves **0/244** theorems at every k, with *or* without skeletons.

- Base outputs `sorry` or echoes the statement in 58.8% of attempts.
- RL model emits `sorry` in $<0.2\%$ of attempts; collapse manifests as failing-but-valid `rw`/`apply` sequences.

RL is the locus of both proof capability and the inference-time collapse that limits it.

Per-skeleton contribution (within $k{=}16$)



24 B-RL $k{=}16$ shards, within-shard counting. Empty skeleton carries most of the load (3.29/4.54 solves per shard); the other 14 skeletons together pick up the long tail — exactly the 17 theorems A-RL i.i.d. misses.

Cross-model generalisation

A→B contrast on two further 7B Lean provers:

- **RL-trained V2-7B:** $+3$ *frontier* theorems no i.i.d. baseline reaches.
- **SFT Goedel-Prover:** $-\mathbf{10.0} \pm \mathbf{4.4}$ theorems ($n{=}3$ seeds). *Helps RL, hurts SFT.*

Distributional evidence of collapse

Across **13,159** stochastic V1.5-RL samples, the median theorem gets only **2** distinct first-tactic heads. **49%** of the benchmark gets a deterministic single strategic opening.

Code, prompts, per-attempt logs: `github.com/zacn04/icml_hints_for_provers`